Muhammad Shams Ghani

Machine Learning Engineer · Computer Vision & Representation Learning

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Status: Valid UK Visa until January 2028 · Available for immediate relocation to the UK (No immediate sponsorship required)

EDUCATION

MSc Artificial Intelligence — Queen's University Belfast, UK — Sep 2024 – Sep 2025

- **MSc Dissertation:** "*Deep FOCUS: Focal Loss Optimisation for Single-Molecule Localisation Microscopy (SMLM) under Severe Sparsity*" (Supervised by Dr. Richard Gault).
- Systematically evaluated four loss functions (L1+L2, MSE, BCE, Focal Loss) across 10,000 synthetic microscopy samples generated with ImageJ ThunderSTORM.
- **Key Finding:** Demonstrated Focal Loss achieved superior reconstruction (33.8 dB PSNR, +13.6 dB vs L1+L2 and +16.7 dB vs MSE baselines) at ~20 nm spatial resolution, establishing that loss landscape geometry matters at least as much as architecture under extreme spatial class imbalance (>99.5% background).
- **Core Modules:** Deep Learning, Computer Vision, AI for Health, Machine Learning, Knowledge Engineering.

BSc Computer Science — GIK Institute of Engineering Sciences & Technology — Aug 2019 – Jun 2023

- Specialized in Artificial Intelligence, Deep Learning, Data Science, and Information Retrieval.
- **BSc Thesis:** *Text-Conditional Latent Audio Diffusion Platform* with emotion-preserving representation synthesis.

PROFESSIONAL EXPERIENCE

Graduate Trainee (AI & ML Engineering) — Itanz — Oct 2026 – Present

- Fine-tuned dense retrieval encoders (all-MiniLM-L6-v2) via Multiple Negatives Ranking Loss (MNRL) across 57,312 chunks, resolving cross-domain polysemy and improving MRR@10 by +0.13 and Hit Rate@1 to >80%.
- Engineered an automated Document AI pipeline rendering scanned clinical PDFs to high-DPI payloads for local Vision-Language Models (Qwen3.5) with Pydantic schema validation.
- Conducted GPU-accelerated RAG evaluation benchmarks, establishing optimal Pareto retrieval bounds (K=3) and designing hallucination-safe abstention tripwires.
- Executed production MLOps validation: model packaging, concurrency stress testing, drift detection strategies, and API security auditing.

Machine Learning Engineer — Profundus (Computer Vision Startup, Islamabad) — Jun 2023 – Jun 2024

- Owned end-to-end ML pipeline for an eyewear virtual try-on system from dataset collection to real-time client deployment.
- Engineered neural background segmentation using contrast maps and image segmentation, eliminating manual product tagging.
- Built multimodal feature extraction and automated cataloging systems processing 10,000+ product images.

AI Engineering Intern — Profundus — Jul 2022 – Oct 2022

- Built visual search microservice for Gulzari Optics (10,000+ items) using VGG16 embeddings + Annoy via DeepImageSearch, served via FastAPI (later updated to FAISS).
- Deployed microservice with full interactive documentation, robust error handling, and performance monitoring.

OPEN SOURCE & TECHNICAL RESEARCH

OpenMouse (@openmouse/protocol) — Core Contributor & Protocol Reverse Engineer — 2026

- Reverse-engineered proprietary USB interrupt transfer reports for gaming mice via Wireshark. Authored TypeScript driver specifications and native WebHID loopback bridge routines, enabling driverless mouse configuration directly in-browser.

SCS-1 Reasoning Harness — Experimental Research Prototype — 2026

- Prototyping an external executive supervisory harness for iterative LLM reasoning that tracks semantic drift between reasoning steps, checks directional NLI entailment to decide when to halt, and applies a temperature ramp when the

model enters rumination loops.

LEADERSHIP & EXTRACURRICULAR

Director of External Operations & Senior Consultant — **GIK Consulting Group** Feb 2021 – Feb 2023

- Resolved complex engineering case study for Y Combinator-backed startup Byte (pro bono). Designed analytical training frameworks that improved junior consultant performance metrics by 50%.

President — **Microsoft Learn Student Ambassadors Club** 2021 – 2023

- Led technical training workshops on cloud computing and Python for 100+ students at GIKI.

TECHNICAL CAPABILITIES

Languages:	Python (Expert), TypeScript, JavaScript, C++, SQL
Deep Learning & Vision:	PyTorch (CUDA), U-Net, Diffusion Models, Vision-Language Models (VLM), OpenCV, scikit-image, TensorFlow
Retrieval & NLP:	Contrastive Learning (MNRL), SentenceTransformers, FAISS, PEFT / LoRA (Qwen), RAG Evaluation, Hugging Face
Systems & Tools:	Docker, FastAPI, WebHID API, Linux / Bash, Git, ImageJ / ThunderSTORM, LaTeX